# 1 The logistic sigmoid

**(a)** Compute and simplify the derivative of the binary logistic sigmoid. (1 pt)

**(b)** Show that the tanh function $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, another function sometimes used as the activation function in neural networks, is simply a scaled and shifted version of the binary logistic sigmoid. (2 pts)

**(c)** Consider binary classification: Given two points of class 1, one at $(1,1)$ and one at $(2,2)$ as well as two points of class 2, at $(1,2)$ and $(2,3)$, find a weight vector $\mathbf{w}$ and bias $b$ such that the activation

$$a = \sigma(\mathbf{w}^T\mathbf{x} + b)$$

separates the two classes. (2 pts)

a) $\sigma(\beta^T x) = \frac{1}{1+\exp(-\beta^T x)} = (1+\exp(-\tau))^{-1}$ $\qquad \beta^T x = \tau$

$$\frac{d\sigma}{d\tau} = -1\,(1+\exp(-\tau))^{-2} \cdot (-\exp(-\tau))$$

$$= \frac{\exp(-\tau)}{(1+\exp(-\tau))^2} = \frac{\exp(-\beta^T x)}{(1+\exp(-\beta^T x))^2} \cdot \frac{\partial \tau}{\partial x}$$

$$= \frac{\exp(-\beta^T x)}{(1+\exp(-\beta^T x))^2} \cdot \beta^T = (\sigma(\beta^T x) - \sigma(\beta^T x)^2) \cdot \beta^T = \beta^T \sigma(\beta^T x)(1-\sigma(\beta^T x))$$

b) $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ $\qquad e^x \cdot e^{-x} = e^0 = 1$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

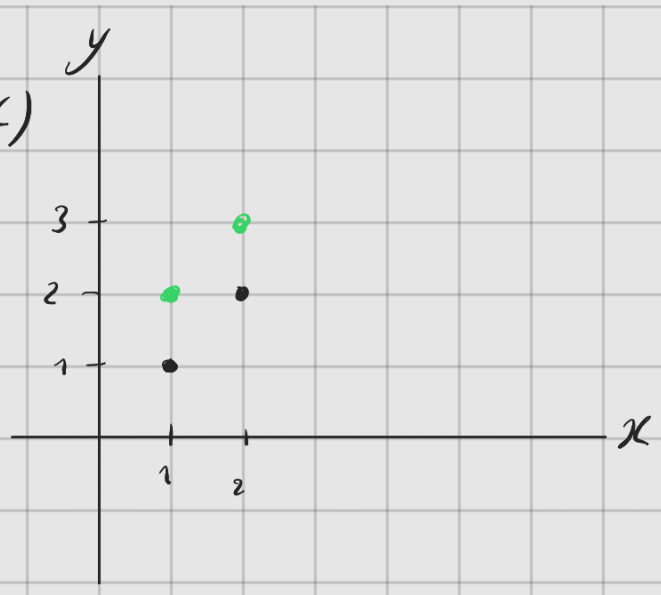
$$\rightarrow \sigma(2x) = \frac{1}{1+e^{-2x}} \quad | \cdot 2$$

$$= \frac{2}{1+e^{-2x}} \quad | -1$$

$$2\sigma(2x) - 1 = \frac{1-e^{-2x}}{1+e^{-2x}} = \frac{e^x \cdot e^{-x} - e^{-x} \cdot e^{-x}}{e^x \cdot e^{-x} + e^{-x} \cdot e^{-x}} = \frac{e^{-x}(e^x - e^{-x})}{e^{-x}(e^x + e^{-x})} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \tanh(x)$$

$$\tanh(x) = 2\sigma(2x) - 1$$

c)




• class 1

• class 2

$$a = \sigma(w^T x + b) = \frac{1}{1 + e^{-(w^T x + b)}}$$

for class 1 $\sigma(z) \approx 1$

for class 2 $\sigma(z) \approx 0$

$w^T x + b \gg 0$

$w^T x + b \ll 0$

$$\left.\begin{array}{l} w_1 \cdot 1 + w_2 \cdot 1 + b > 0 \\ w_1 \cdot 2 + w_2 \cdot 2 + b > 0 \\ w_1 \cdot 1 + w_2 \cdot 2 + b < 0 \\ w_1 \cdot 2 + w_2 \cdot 3 + b < 0 \end{array}\right\} \left.\begin{array}{l} w_1 > 0 \\ w_2 < 0 \end{array}\right\} w_1 = -w_2 \quad \left.\begin{array}{l} b > 0 \end{array}\right\} \text{let } \begin{array}{l} w_1 = 1 \\ w_2 = -1 \\ b = 1 \end{array}$$

$$\left.\begin{array}{l} 1 - 1 + 1 > 0 \\ 2 - 2 + 1 > 0 \\ 1 - 2 + 1 = 0 \\ 2 - 3 + 1 = 0 \end{array}\right\} \text{let } b = 0{,}5 \qquad \begin{array}{l} 0{,}5 > 0 \\ 0{,}5 > 0 \\ -0{,}5 < 0 \\ -0{,}5 < 0 \end{array} \qquad \begin{array}{l} w = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ b = 0{,}5 \end{array}$$

$$a = \sigma(x_1 - x_2 + 0{,}5)$$

this activation function is able to separate the two classes

## 3 Log-sum-exp and soft(arg)max

The log-sum-exp and soft(arg)max[1] functions are defined on vectors $\boldsymbol{\sigma} \in \mathbb{R}^k$ as

$$\mathrm{lse}(\boldsymbol{\sigma}; \lambda) = \frac{1}{\lambda} \log \left( \sum_{j=1}^{K} \exp(\lambda \sigma_j)) \right) \quad \text{and} \quad \mathrm{soft(arg)max}(\boldsymbol{\sigma}; \lambda)_k = \frac{\exp(\lambda \sigma_k)}{\sum_{j=1}^{K} \exp(\lambda \sigma_j))},$$

with a scalar parameter $\lambda \in \mathbb{R}^+$ and $k = 1, .., K$.

**(a)** On which subset of the vectors $\boldsymbol{\sigma}^1 = (1, 2, 3)^T$, $\boldsymbol{\sigma}^2 = (11, 12, 13)^T$, $\boldsymbol{\sigma}^3 = (10, 20, 30)^T$ does the soft(arg)max yield identical results? Show in general whether the soft(arg)max is invariant under (i) constant offset and (ii) rescaling of its input. (2 pts)

**(b)** Make a 2D contour plot of $\mathrm{lse}((\sigma_1, \sigma_2)^T; \lambda)$ for $\lambda \in \{1, 10, 100\}$ and both $\sigma_1$ and $\sigma_2$ in the range $[-1, 1]$. Compare this to a contour plot of $\max(\sigma_1, \sigma_2)$ over the same range. (2 pts)

**(c)** Plot the two components of the soft(arg)max, over the same range and the same choices for $\lambda$ as in (c), but as images instead of contour plots. Compare this to corresponding plots of the two components of the arg max over the 2D vectors, represented as a 2D one-hot vectors instead of indices:

$$\mathrm{onehot}(\arg\max_i \sigma_i) = \begin{cases} (1, 0)^T & \text{if } \sigma_1 > \sigma_2, \\ (0, 1)^T & \text{else.} \end{cases}$$

(2 pts)

**(d)** Prove that the derivative of the lse is the soft(arg)max. (1 pt)

**(e)** **Bonus** Prove that

$$\lim_{\lambda \to \infty} \mathrm{lse}(\boldsymbol{\sigma}; \lambda) = \max(\boldsymbol{\sigma}),$$

for all $\boldsymbol{\sigma} \in \mathbb{R}^n$. (2 pts)

[1] In the literature known as just "softmax".

a)

$$\mathrm{soft(arg)max}(\sigma_1)_1 = \frac{e^1}{e^1+e^2+e^3} \approx 0{,}09$$

$$\mathrm{soft(arg)max}(\sigma_1)_2 = \frac{e^2}{e^1+e^2+e^3} \approx 0{,}24$$

$$\mathrm{soft(arg)max}(\sigma_1)_3 = \frac{e^3}{e^1+e^2+e^3} \approx 0{,}66$$

$$\mathrm{soft(arg)max}(\sigma_2)_1 = \frac{e^{11}}{e^{11}+e^{12}+e^{13}} = 0{,}09$$

$$\mathrm{soft(arg)max}(\sigma_2)_2 = \frac{e^{12}}{e^{11}+e^{12}+e^{13}} = 0{,}24$$

$$\mathrm{soft(arg)max}(\sigma_2)_3 = \frac{e^{13}}{e^{11}+e^{12}+e^{13}} = 0{,}66$$

The soft(arg)max is same for the $\sigma^1$ and $\sigma^2$

$$\text{Soft(arg)max}(\sigma_3)_1 = \frac{e^{10}}{e^{10}+e^{20}+e^{30}} \approx 0$$

$$\text{Soft(arg)max}(\sigma_3)_2 = \frac{e^{20}}{e^{10}+e^{20}+e^{30}} \approx 0$$

$$\text{Soft(arg)max}(\sigma_3)_3 = \frac{e^{30}}{e^{10}+e^{20}+e^{30}} \approx 1$$

$\alpha$ = constant

i) In general

$$\text{Soft(arg)max}(\sigma+\alpha)_k = \frac{e^{(\sigma_k+\alpha)}}{\sum_{i=1}^{k} e^{(\sigma_i+\alpha)}} = \frac{e^{\alpha}\cdot e^{\sigma_k}}{\sum_{i=1}^{k} e^{\alpha}\cdot e^{\sigma_i}} = \underbrace{\frac{e^{\alpha}}{e^{\alpha}}}_{1}\cdot\frac{e^{\sigma_k}}{\sum_{i=1}^{k} e^{\sigma_i}} \checkmark$$

ii) $$\text{Soft(arg)max}(\sigma;\lambda) = \frac{e^{\lambda\cdot\sigma_k}}{\sum_{j=1}^{k} e^{\lambda\cdot\sigma_j}}$$

, the $\lambda$ can't be factored out. The Soft(arg)max will therefore be different for different $\lambda$.

<u>example</u>

$\lambda = 1$ $\quad \text{Soft(arg)max}(\sigma;1) = \frac{e^{\sigma_k}}{\sum_{i=1}^{k} e^{\sigma_i}}$

$\lambda = 10$

$$\text{Soft(arg)max}(\sigma;10) = \frac{e^{10\sigma_k}}{\sum_{i=1}^{k} e^{10\sigma_i}} = \frac{(e^{\sigma_k})^{10}}{\sum_{i=1}^{k} (e^{\sigma_i})^{10}} \neq \frac{e^{\sigma_k}}{\sum_{i=1}^{k} e^{\sigma_i}} \quad !$$

b) look at the Jupyter file

$$\frac{\partial}{\partial\sigma_k} e^{\sigma_k} = e^{\sigma_k}$$

c) look at the Jupyter file

$$\frac{\partial}{\partial\sigma_k} e^{\sigma_i} = 0$$

d) $$\text{lse}(\sigma) = \log\left(\sum_{i=1}^{k} e^{\sigma_i}\right)$$

$$\frac{d\,\text{lse}}{d\sigma_k} = \underbrace{\frac{1}{\sum_{i=1}^{k} e^{\sigma_i}}}_{\text{outer product}} \cdot \underbrace{\frac{\partial}{\partial\sigma_k}\left(e^{\sigma_i}+e^{\sigma_{i+1}}+\dots+e^{\sigma_k}\right)}_{\text{inner product}} = \frac{e^{\sigma_k}}{\sum_{i=1}^{k} e^{\sigma_i}} = \text{Soft(arg)max}(\sigma)_k$$

$$\mathcal{L}_{se}(\sigma;\lambda) = \frac{1}{\lambda}\log\left(\sum_{i=1}^{k} e^{\lambda\sigma_i}\right)$$

$$\frac{d}{d\sigma_k}\mathcal{L}_{se}(\sigma;\lambda) = \frac{\lambda e^{\lambda\sigma_k}}{\lambda\cdot\sum_{i=1}^{k} e^{\lambda\sigma_i}} = \frac{e^{\lambda\sigma_k}}{\sum_{i=1}^{k} e^{\lambda\sigma_i}} = \mathrm{softargmax}(\sigma;\lambda)_k$$

e)

$$\lim_{\lambda\to\infty}\mathcal{L}_{se}(\sigma;\lambda) = \max(\sigma) \quad \forall\sigma\in\mathbb{R}^n$$

$\log(a\cdot b) = \log(a)+\log(b)$

$\sigma_m$ - is one of the $\sigma_i$ for an arbitrary $i$

$\sigma_m \geq \sigma_i$ for $i\in\{1,\dots,k\}$

$\max(\sigma) = \sigma_m$

$$\frac{1}{\lambda}\log\left(\sum_i^k e^{\lambda\sigma_i}\right) = \frac{1}{\lambda}\log\left(e^{\lambda\sigma_i}+e^{\lambda\sigma_{i+1}}+\dots+e^{\lambda\sigma_m}+\dots+e^{\lambda\sigma_k}\right)$$

$$= \frac{1}{\lambda}\log\left(e^{\lambda\sigma_m}\cdot\left(e^{\lambda(\sigma_i-\sigma_m)}+e^{\lambda(\sigma_{i+1}-\sigma_m)}+\dots+1+\dots+e^{\lambda(\sigma_k-\sigma_m)}\right)\right)$$

$$= \frac{1}{\lambda}\log\left(e^{\lambda\sigma_m}\cdot\left(\sum_i^k e^{\lambda(\sigma_i-\sigma_m)}\right)\right)$$

$$= \frac{1}{\lambda}\log\left(e^{\lambda\sigma_m}\right) + \frac{1}{\lambda}\log\left(\sum_i^k e^{\lambda(\sigma_i-\sigma_m)}\right)$$

$$= \frac{\lambda\sigma_m}{\lambda} + \frac{1}{\lambda}\log\left(\sum_i^k e^{\lambda(\sigma_i-\sigma_m)}\right)$$

$$e^{\sigma_i-\sigma_m} = \begin{cases} 1, & \sigma_i = \sigma_m \\ 0, & \sigma_i < \sigma_m \end{cases}$$

for $\lambda\to\infty$

$$\lim_{\lambda\to\infty}\mathcal{L}_{se}(\sigma;\lambda) = \sigma_m + \frac{1}{\lambda}\underbrace{\log(1+0+0+\dots)}_{\log(1)=0} = \sigma_m = \max(\sigma)$$

$\square$

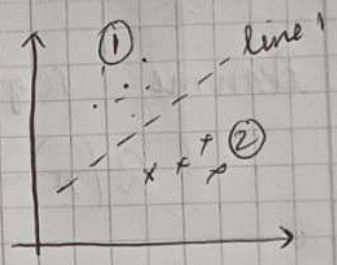
Bonus Question:-

Given

- 2D input
- 1 hidden layer
- ReLU activation for each neuron

Assume linear regions formed ~~R~~ with $n$ neurons be $R(n)$

if $n = 1$, it is given that its is representing as a line in 2D plane




$\therefore R(1) = 2$

if we add '$n$' neurons then

$$R(n) = R(1) + 2 + 3 + \cdots n$$

$\therefore$ Recursively,

$$\Rightarrow \boxed{R(n) = R(n-1) + n}$$

Alternatively,

$$R(n) = R(1) + 2 + 3 + \cdots n$$

$$= \cancel{R}\, 2 + (2 + 3 + \cdots n) \quad [R(1) = 2 \text{ regions}]$$

$$= 2 + \sum_{i=2}^{n} i$$

$$= 2 + \sum_{i=1}^{n} i - 1 = 2 + \left(\frac{n(n+1)}{2} - 1\right)$$

$$= 1 + \frac{n(n+1)}{2}$$

$$\therefore \boxed{R(n) = 1 + \frac{n(n+1)}{2}}$$